

Anil Kumar

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EDUCATION

Arizona State University: Masters of Science, Computer Science, 4.0|4.0

National Institute of Technology Hamirpur: Bachelors of Technology, Computer Science, 3.32|4.0

EXPERIENCE

- Software Engineer, Google, Mountain View, CA

Jan 2026

 - Building youtube channel relatedness information graphs to detect platform abusers in real time.
 - Dynamic graph building using user’s behavioral, technical, video, commenter features to counter the adversarial view rate on youtube.
- Software Engineer, Google, Durham, NC

July 2025 - Jan 2026

 - Integrated Object storage API with airgapped google distributed hosted cloud services.
 - Build and optimize S3-compatible APIs to ensure seamless data access, low-latency performance, and high throughput for diverse customer workloads.
 - Developed end to end testing for GDCH integration with AWS and Azure object storage APIs.
- Software Engineer, Google, San Bruno, CA

Jan 2023 - July 2025

 - Built youtube channel-to-channel relatedness information graph to develop relationships among youtube active and suspended channels.
 - Built YouTube trust and safety pipeline to model neighborhood graphs and enforce actions against bad actors.
 - Developed a ground truth collection library to support online and offline clustering and classifier training, preventing engagement abuse, resulting in the termination of 30M abusive accounts within six months.
 - Reduced channel enforcement time from 24 hours to real time to keep violative views rate lower and got youtube spot bonus award.
 - Built appeal serving backend for youtube accounts and behavioral abuse related suspended channels.
- Graduate Research Assistant, Arizona State University

Jan 2021 – Dec 2022

 - Designed and implemented a performance simulation system for Hyperledger Fabric-based private blockchain networks using Petri net models, analyzing behavior under cyberattack scenarios.
 - Built an IoT identity management system to mirror device digital twins over Hyperledger Fabric, performing real-time anomaly detection with machine learning models using historical data, hosted on Google Cloud Platform.
 - Created and delivered coursework for Engineering Blockchain Applications for a class of 120 students as a Teaching Assistant.
- Software Engineering Intern, Google, San Bruno, CA

May 2022–Aug 2022

 - Designed a file size estimation system in C++ for YouTube's data warehouse job scheduling, optimizing network resource allocation for internal teams.
 - Generated actionable data insights through Spanner updates in C++ to analyze network resource consumption for replication tasks on the Colossus file system, identifying a 10x excess in daily replication.
 - Enhanced the replication job algorithm in the Colossus file system by optimizing the slice size metric for job scheduling, improving resource efficiency.
- Software Engineer, Samsung Research, India

July 2017 – Dec 2020

 - Developed India’s leading video and music streaming applications, **Hotstar** and **Jio Cinema**, for Smart TVs using React and JavaScript, achieving over 50M active users.
 - Integrated Samsung Tizen TV applications with voice assistant **Bixby**, enabling extended features through API integrations between the application and platform layers.
 - Built a web application mapped to a channel number on Tizen TV OS using the **TV Plus framework**, with the prototype showcased globally at the **Samsung Developer Conference 2018**, San Francisco.

PATENT

- A method for adaptive action learning for unstructured utterances in voice assistants ([US20220148580A1](#))
- Detected the most relevant action for the given unstructured utterance by extracting voice features using a Dynamic Bayesian Network.
 - Performed reinforcement learning to update the global model to perform the most relevant action.

PUBLICATIONS

- [“Joint Modeling of Hyperledger Fabric and Sybil attack: Petri Net approach”](#), Simulation Modeling Practice and Theory, Oct 2022.

CONFERENCE

[Applied Computer Security Associates Conference 2022](#)

- Engineered a [decentralized Identity and Access Management](#) (IAM) framework using Hyperledger Fabric to eliminate single points of failure in centralized IoT systems, ensuring secure peer-to-peer data exchange and shared resource ownership and integrated real-time security monitoring
- Developed an automated identity generation system using smart contracts that utilize dynamic state parameters to create unique, immutable device IDs and manage secure ownership transfers across the device lifecycle.

PROJECTS

- Developed a block explorer for Accumulate using React, Google Firebase, and Accumulate test net APIs that helps users to look into historical blockchain data and won the Accumulate blockchain 2022 hackathon on HackerEarth.
- [Accumulate Blockchain Hackathon 2022 winner](#), built block explorer for Accumulate blockchain.

SKILLS

- **Programming languages:** C++, Python, Go, JavaScript, Java, SQL, Rust, MySQL, PostgreSQL
- **Technologies/Frameworks/Functional Skills:** Docker, Kubernetes, NodeJS, React, jQuery, HTML5, CSS, Tizen SDK, REST API, HELM, Google Cloud Platform, AWS, Jira, Hadoop, Spark, Hyperledger fabric